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Three-dimensional kinematics and ground reaction forces during the instep and outstep soccer kicks in pubertal players

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Abstract

The purpose of the present study was to compare the three-dimensional kinematics of the lower extremities and ground reaction forces between the instep kick and the kick with the outside area of the foot (outstep kick) in pubertal soccer players. Ten pubertal soccer players performed consecutive kicking trials in random order after a two-step angled approach with the instep and the outstep portion of the foot. Three-dimensional data and ground reaction forces were measured during kicking. Paired t-tests indicated significantly higher ($P < 0.05$) ball speeds and ball/foot speed ratios for the instep kick compared with the outstep kick. Non-significant differences in angular and linear sagittal plane kinematic parameters, temporal characteristics, and ground reaction forces between the instep and outstep soccer kicks were observed ($P > 0.05$). In contrast, analysis of variance indicated that the outstep kick displayed higher hip internal rotation and abduction, knee internal rotation, and ankle inversion than the instep kick ($P < 0.05$). Our results suggest that the instep kick is more powerful than the outstep kick and that different types of kick require different types of skill training.

Keywords: *Instep kick, outstep kick, kinematics, ball speed, ground reaction forces*

Introduction

Several types of kick are performed during a soccer game, such as the drop kick, the punt kick, the instep kick, the kick with the inside and the outside portion of the foot (outstep) (Lees & Nolan, 1998). Of these, the kinematic, kinetic, and electromyographic characteristics of instep soccer kicks have been most extensively examined (Barfield, 1995; De Proft, Clarys, Bollens, Cabri, & Dufour, 1988; Kellis & Katis, 2007; Rodano & Tavana, 1993).

A few studies have compared the biomechanical characteristics of different types of kick (Nunome, Asai, Ikegami, & Sakurai, 2002; Wang & Griffin, 1997). For example, the inside kick, which is mainly used when greater accuracy is required, displays lower ball speeds and a lower hip rotation torque and angular velocity compared with the instep kick (Nunome et al., 2002). The inside kick, frequently reported as place or curve kick, displays higher spin and lift of the ball (Asai, Carre, Akatsuka, & Haake, 2002; Carre, Asai, Akatsuka, & Haake, 2002) and lower angular momentum compared with the instep kick (Wang & Griffin, 1997).

Kicking with the outside portion of the foot (outstep kick) is another frequent type of soccer kick. Outstep kicks, similarly to instep kicks, are characterized by a multi-step approach and increased ball velocity and are used when the aim of the player is to bend the ball around an opponent or to kick the ball around a defensive wall. However, the ball is hit with the outer part of the foot. In theory, if the position of the foot and the surface of collision on ball impact differ between the two kicking types, then the kinematic patterns of the kicking leg should also differ, as the player must bring the kicking foot into the appropriate position so that the outer area of the foot contacts the ball. This final foot position should be the result of sequential movements of the foot, the tibia, and the femur in more than one plane, at least during the final phase of the kick (near ball contact). If this is the case, the outstep kick should display

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different kinematic patterns compared with the instep kick. In addition, the surface of collision during the outstep kick is smaller compared with that of the instep kick. This suggests that the quality of ball–foot contact might differ between the two kicks. Consequently, one might expect that biomechanical indices of soccer kicking, such as maximal ball speed, final foot speed, as well as the ball speed/foot speed ratio (Kellis & Katis, 2007; Lees & Nolan, 1998), might differ between the two kicks.

The temporal characteristics of the kick as well as support foot placement are also important aspects of a soccer kick. For example, kicks performed with the same technique but for a different total duration might not have the same result, in terms of final ball speed and accuracy (Opavsky, 1988). Similarly, in theory, altered support foot placement may result in altered ground reaction forces underneath the support foot, which, in turn, might affect soccer kick performance (Kellis, Katis, & Gissis, 2004). Consequently, when comparing two different types of kick, differences in temporal and ground reaction force characteristics should also be examined.

A good kicking technique is a fundamental skill of every soccer player. For this reason, the fundamental aspects of kicking are being taught from an early age. Soccer training for young players includes skills and drills for all types of kicks (Weineck, 1997). A few studies have examined the biomechanics of soccer kicking in younger players (Luhtanen, 1988). For example, Luhtanen (1988) reported ball speeds for young players ranging from 14.9 $m \cdot s^{-1}$ at the age of 10 years to 22.2 $m \cdot s^{-1}$ at the age of 17 years, while ball velocity can exceed 30 $m \cdot s^{-1}$ among adult male soccer players (Asami & Nolte, 1983; Opavsky, 1988; Rodano & Tavana, 1993). All these values, however, refer to instep soccer kick performance. To our knowledge, outstep kick kinematics in young soccer players have not been investigated. If outstep kicks display different kinematic characteristics compared with instep kicks, then this indicates that each kick requires different muscle and joint coordination strategies and therefore different types of skill training.

The purpose of the present study was to compare the three-dimensional kinematics of the lower extremities and ground reaction forces between the instep kick and the kick with the outside portion of the foot in pubertal soccer players. The main hypotheses tested were that: (a) kicking with the outside portion of the foot would display lower ball speeds compared with the instep kick, (b) the outstep kick would demonstrate different kinematic patterns of the swinging limb compared with the instep kick, and (c) the temporal characteristics and ground reaction forces would also differ between the two types of kick.

Methods

Participants

Ten skilled pubertal soccer players (age 13.6 ± 0.7 years, mass 47.5 ± 13.1 kg, height 1.54 ± 0.09 m) volunteered to participate in the present study. The players had a minimum of 5 years of experience, with a training frequency of at least twice a week plus one game per week, and no history of injury within the 6 months prior to the final measurements. Seven players preferred to kick with their right leg and three preferred to kick with their left leg. Informed written consent was received from the participants and their parents before testing began. The university ethics committee approved the protocol.

Test procedure

A warm-up consisting of 10 min submaximal running on a treadmill and several stretching exercises for the whole body was conducted before the measurements. Kicks were performed against the centre of a goal located 7 m from the ball. A standard sized and inflated ball (FIFA approved) was used for the measurements, while participants wore soccer boots. In the main session, each participant was instructed to kick the ball, as hard as possible, using a two-step angled (45°) approach relative to the target. Five trials were performed with the instep and five trials with the outstep area of the foot. The trial that displayed the highest ball speed was used for further analysis.

Kinematics

Kinematic data were collected with a passive, six-camera, 3D Vicon motion analysis system (Oxford Metrics Ltd., Oxford, UK) at 200 Hz. The cameras were calibrated to a volume of 2.0 m^3 and calibration errors were all below 3 mm.

Reflective spherical markers were placed on selected anatomical landmarks of both limbs to identify segments and joints of the lower extremities: the head of the fifth metatarsal, the heel, mid-shank, lateral malleolus, femoral epicondyle, mid-thigh, greater trochanter, anterior superior iliac spine of the left leg, as well as the sacrum, right and left shoulders. Four additional markers were placed on the surface of the ball. Before each kicking trial, a standing trial was recorded to establish initial joint angle conditions. Marker position was automatically tracked using the Polygon software (Oxford Metrics Ltd., Oxford, UK). Three-dimensional marker position coordinates were computed using the direct linear transformation (DLT) method. The three-dimensional coordinates were expressed as a right-handed orthogonal reference frame, in which the Z

axis was vertical and pointed upwards, the Y axis was horizontal, and the X axis was perpendicular to Z and Y. The resulting displacement–time data of each marker were filtered using a low-pass, fourth-order Butterworth dual-pass filter. Optimal cut-off frequencies were chosen by comparing the residuals of the difference between filtered and unfiltered signals at several cut-off frequencies. The cut-off frequency for filtering the kinematic data was set at 6 Hz.

Angles between the segments were examined by viewing the Euler angles decomposed from the rotation matrix, which describes the orientation of one segment to another. The Euler angle decomposition order was based on an algorithm that extracts the rotation about the Y axis, then the rotation about the new orientation of the X axis, and finally the rotation about the twice-rotated orientation of the Z axis. The axis system of the segments was: X axis in the anteroposterior direction, Y axis in the mediolateral direction, and Z axis in the proximal–distal direction. Therefore, in the present study, the decomposition of the Euler angles for each segment was as follows: the ankle was separated into plantar flexion/dorsiflexion and inversion/eversion angles; the knee was separated into flexion/extension and internal/external rotation angles; and the hip was separated into internal/external rotation and abduction/adduction angles.

Ground reaction forces

The vertical, anteroposterior, and mediolateral components of the ground reaction forces were measured at a sampling frequency of 1000 Hz using a Bertec force plate (Type 4060A, Bertec Corp., Worthington, OH). The force plate was located in the middle of a 5-m long pathway, flush with the floor and concealed with plastic turf to avoid disorientation of the player by focusing on stepping inside the force plate when kicking. The force plate signals were A/D converted and recorded. Subsequently, they were analysed simultaneously with the kinematic data.

Data analysis

To simplify the results, kicking motion was divided in two phases: from the toe-off of the kicking leg to ground contact of the support leg (pre-support phase) and from ground contact to ball impact (support phase). Each phase was set as 100%. Subsequently, the segmental angular velocities and joint displacements were averaged for every 10% of each phase. In addition to the time-series calculation, the following variables were examined for each type of kick: duration of the kick and its phases; maximum ball speed; ball speed/foot (ankle) speed ratio; maximum angular velocity of the hip, the knee, and the ankle; and maximum ground reaction forces. These parameters were selected based on previous studies (Barfield, 1995; Kellis et al., 2004), as they are frequently considered to be important biomechanical indicators of soccer kick performance.

Statistical analysis

Differences between instep and outstep kicks in maximum ground reaction force, ball speed, ball speed/foot speed ratio, kicking duration, and maximum angular velocity of the hip, the knee, and the ankle were examined using paired two-tailed Student's *t*-tests. A two-way analysis of variance (ANOVA) with repeated measures with two within-participant variables (kick, phase) was used to examine differences in angular joint displacements and segmental velocities of the pre-support phase between the two kicks (instep and outstep kick). A separate two-way ANOVA with repeated measures was also used to examine the effects of kick and phase on angular joint displacements and segmental velocities of the support phase. Significant interactions were followed up with simple effects tests and, if significant, *post-hoc* Tukey tests were applied to examine significant differences between pairs of means. Statistical significance was set at $P < 0.05$.

Results

Biomechanical indicators of kicking success

The mean ball speed of the instep kick (19.62 ± 1.89 m · s^{-1}) was significantly higher than that of the outstep kick (18.10 ± 1.49 m · s^{-1}) (Table I, $P < 0.05$). The *t*-test indicated that the ball speed/

Table I. Selected sagittal plane kinematic variables during instep and outstep kicks (mean ± *s*).

	Instep	Outstep
Ball velocity (m · s^{-1})	19.62 ± 1.89	18.10 ± 1.49*
Ball speed/foot speed ratio	1.61 ± 0.41	1.48 ± 0.32*
Maximum angular velocity (deg · s^{-1})		
Hip	774.35 ± 79.31	753.55 ± 71.65
Knee	1625.62 ± 160.81	1648.63 ± 128.71
Ankle	1765.83 ± 162.38	1717.85 ± 229.54
Maximum linear velocity (m · s^{-1})		
Hip	2.69 ± 0.32	2.66 ± 0.40
Knee	7.52 ± 0.43	7.50 ± 0.50
Ankle	12.19 ± 0.51	12.17 ± 1.42
Pre-support phase duration (ms)	519 ± 154	536 ± 176
Support phase duration (ms)	213 ± 69	219 ± 66

*Significant difference between kicks.

foot speed ratio of the instep kick was significantly higher than that of the outstep kick (Table I, $P < 0.05$).

Segmental velocity–phase curves

The sagittal plane hip, knee, and ankle angular velocities for the two phases of the kick are presented in Figure 1. For all ANOVA comparisons, the main effect for type of kick was not significant ($P > 0.05$). No statistically significant interaction effect (type of kick × phase) was observed for hip or knee angular velocity (Figure 1, $P > 0.05$).

The ANOVA results indicated a non-significant interaction (type of kick × phase) for ankle angular velocity during the pre-support phase (Figure 1, $P < 0.05$). In contrast, there was a significant interaction effect during the support phase (Figure 1, $P < 0.05$). *Post hoc* analysis indicated that the instep kick showed significantly higher ankle angular velocity from 40% to 80% of the support phase compared with the outstep kick ($P < 0.05$).

Joint displacement–phase curves

The hip angular displacement curves are presented in Figure 2. The ANOVA showed a significant interaction effect for hip abduction angle during the support phase (Figure 2, $P < 0.05$). *Post hoc* analysis indicated that from 60% to ball impact (100%), greater hip abduction was observed during the outstep kick (Figure 2, $P < 0.05$). No other statistically significant effects on hip abduction and hip adduction were observed ($P > 0.05$).

There was a statistically significant interaction effect on hip internal rotation during the support phase (Figure 2, $P < 0.05$). *Post hoc* analysis indicated that from 20% to 50% of the support phase, the outstep kick displayed increased hip internal rotation values compared with the instep kick (Figure 3, $P < 0.05$). Non-significant interaction and main effects were observed for hip external rotation ($P > 0.05$).

No statistically significant interaction or main effects were observed for knee extension, knee

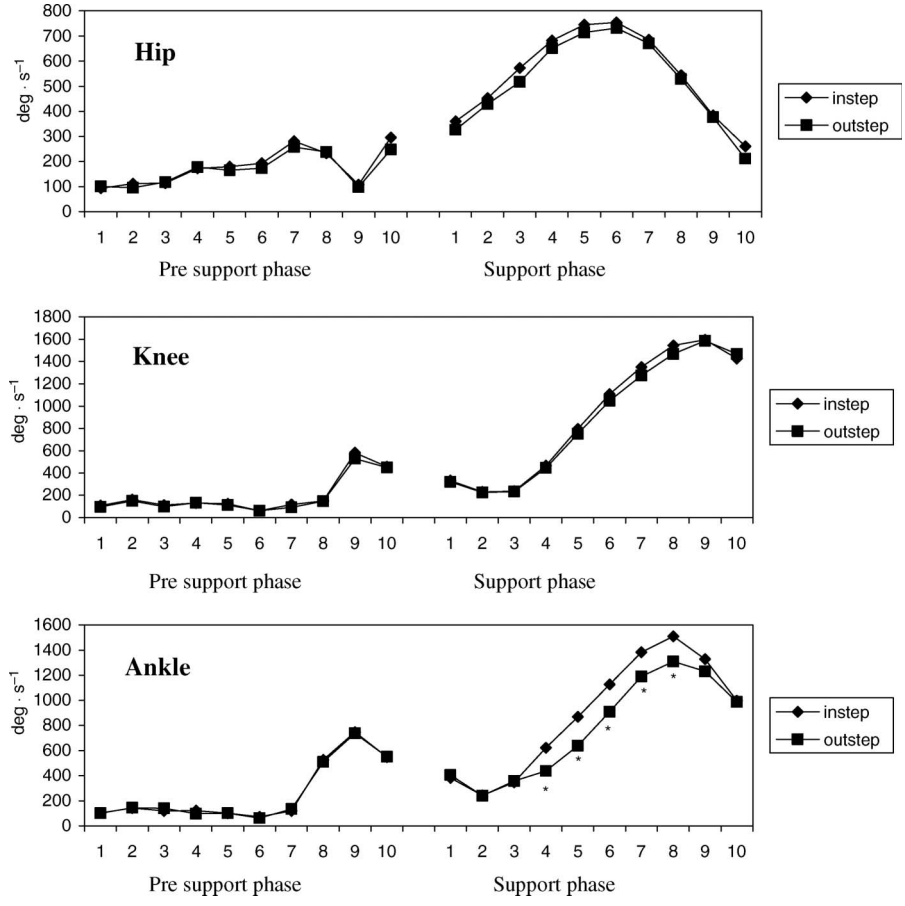

Figure 1. Average segment sagittal angular velocity (deg · s^{-1}) of the swinging leg during the instep and outstep kick. Diagrams on the left indicate values expressed for every 10% from the toe-off of the swinging leg until ground contact (pre-support phase). Diagrams on the right indicate values expressed for every 10% from ground contact to ball impact (support phase) (1: 1–10%; 2: 11–20%; 3: 21–30%; 4: 31–40%; 5: 41–50%; 6: 51–60%; 7: 61–70%; 8: 71–80%; 9: 81–90%; 10: 91–100% of each phase). *Significantly different from the instep kick ($P < 0.05$).

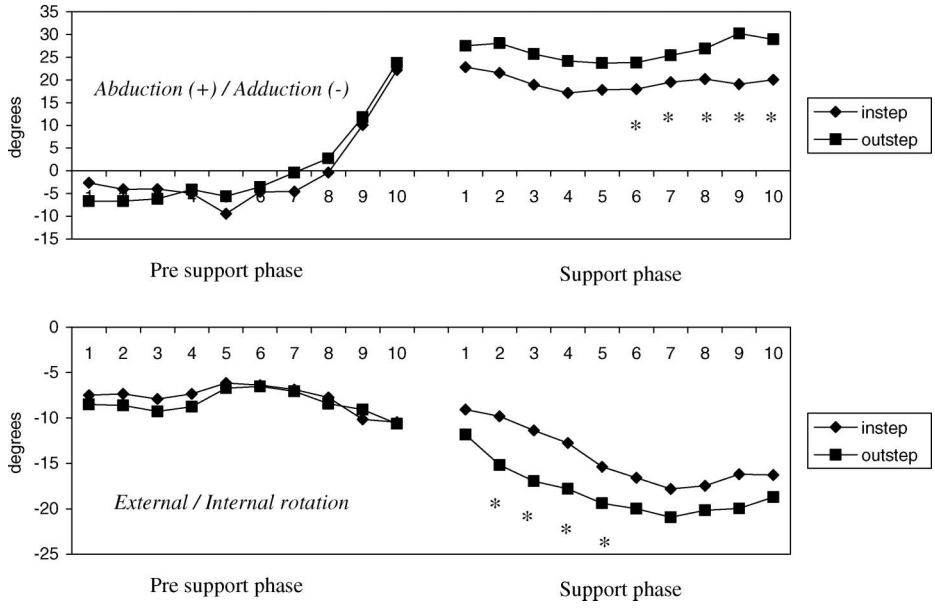

Figure 2. Average hip displacement (degrees) of the swinging leg during the instep and outstep kick. Upper diagram indicates hip adduction/abduction. Lower diagram indicates hip internal/external rotation. Values are expressed for every 10% from the toe-off of the swinging leg until ground contact (pre-support phase) and from ground contact to ball impact (support phase) (1: 1–10%; 2: 11–20%; 3: 21–30%; 4: 31–40%; 5: 41–50%; 6: 51–60%; 7: 61–70%; 8: 71–80%; 9: 81–90%; 10: 91–100% of each phase). *Significantly different from the instep kick ($P < 0.05$).

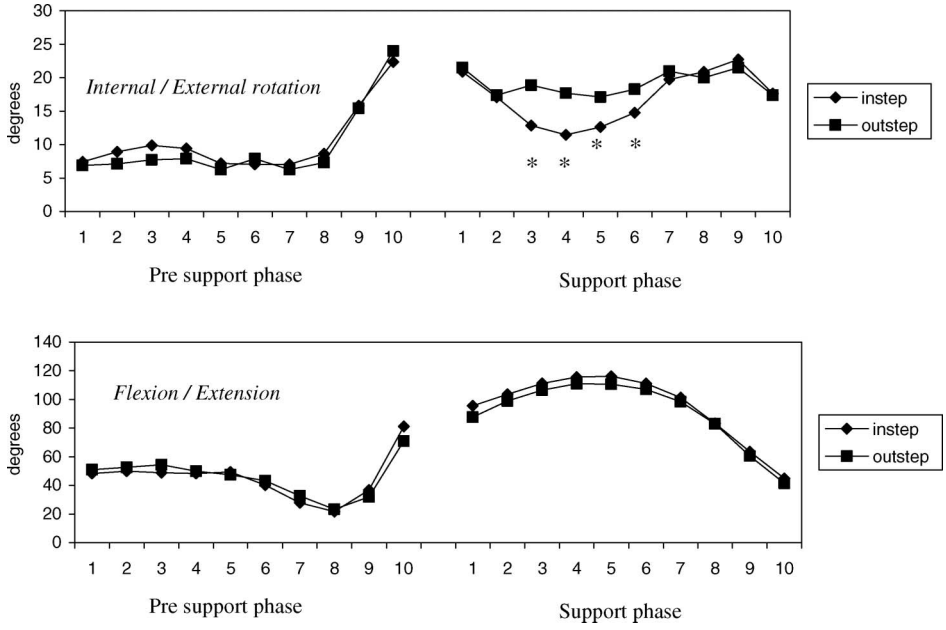

Figure 3. Average knee displacement (degrees) of the swinging leg during the instep and outstep kick. Upper diagram indicates knee internal/external rotation. Lower diagram indicates knee flexion/extension. Values are expressed for every 10% from the toe-off of the swinging leg until ground contact (pre-support phase) and from ground contact to ball impact (support phase) (1: 1–10%; 2: 11–20%; 3: 21–30%; 4: 31–40%; 5: 41–50%; 6: 51–60%; 7: 61–70%; 8: 71–80%; 9: 81–90%; 10: 91–100% of each phase). *Significantly different from the instep kick ($P < 0.05$).

flexion or external rotation angle (Figure 3, $P > 0.05$). In contrast, there was a statistically significant interaction effect for knee internal rotation during the support phase (Figure 3, $P < 0.05$). *Post hoc* analysis indicated that from 30% to 60% of the support phase, a higher internal rotation was observed during the outstep kick compared with the instep kick (Figure 3, $P < 0.05$).

The ANOVA showed non-statistically significant interaction and main effects for ankle plantar flexion, dorsiflexion, and eversion angles (Figure 3, $P > 0.05$). In contrast, during the support phase, a

statistically significant interaction was observed for ankle inversion angle (Figure 4, $P < 0.05$). *Post hoc* analysis indicated a higher ankle inversion during the outstep kick compared with the instep kick from the beginning (10%) to 70% of the support phase (Figure 4, $P < 0.05$).

Temporal parameters of the kick

Table I shows the selected temporal parameters. The *t*-test indicated a non-significant difference in the duration of the pre-support and the support phase between the two kicks ($P > 0.05$).

Ground reaction forces and maximum velocities

The *t*-tests indicated non-significant differences in the vertical, mediolateral, and anteroposterior ground reaction forces between the two kicks (Figure 5, $P > 0.05$). Furthermore, non-significant differences in maximum linear and sagittal plane angular velocities were observed between the two types of kick (Table I, $P > 0.05$).

Discussion

The main findings of the present study are that compared with the instep kick, the outstep kick displayed: (a) lower ball speeds and ball speed/foot speed ratios; (b) higher hip internal rotation and abduction, internal knee rotation, and ankle inversion values but similar sagittal plane kimematics; and (c) similar temporal parameters and ground reaction forces. Therefore, the first and the second hypotheses of the study are accepted and the third hypothesis is rejected.

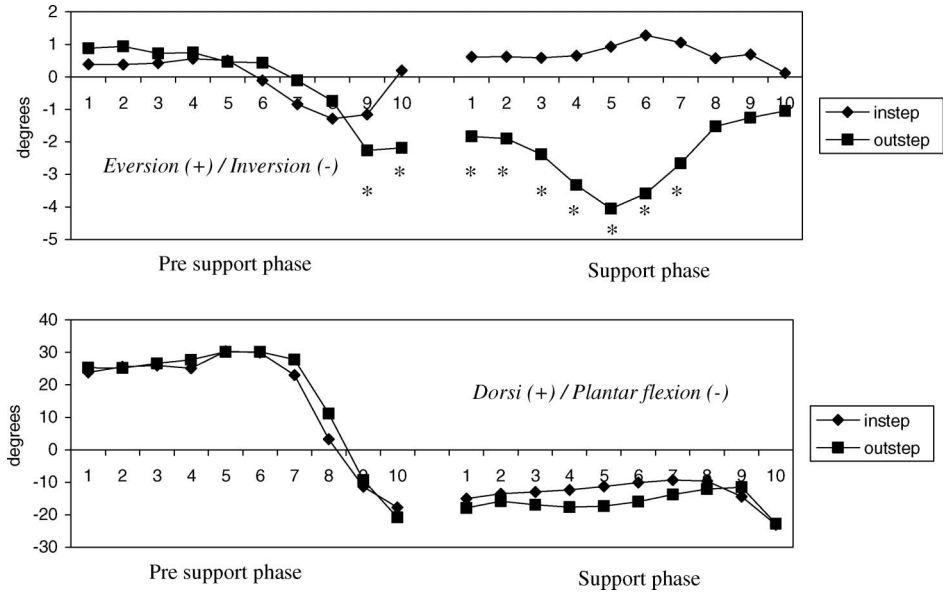

Figure 4. Average ankle displacement (degrees) of the swinging leg during the instep and outstep kick. Upper diagram indicates ankle inversion/eversion. Lower diagram indicates ankle dorsiflexion/plantar flexion. Values are expressed for every 10% from the toe-off of the swinging leg until ground contact (pre-support phase) and from ground contact to ball impact (support phase) (1: 1–10%; 2: 11–20%; 3: 21–30%; 4: 31–40%; 5: 41–50%; 6: 51–60%; 7: 61–70%; 8: 71–80%; 9: 81–90%; 10: 91–100% of each phase). *Significantly different from the instep kick ($P < 0.05$).

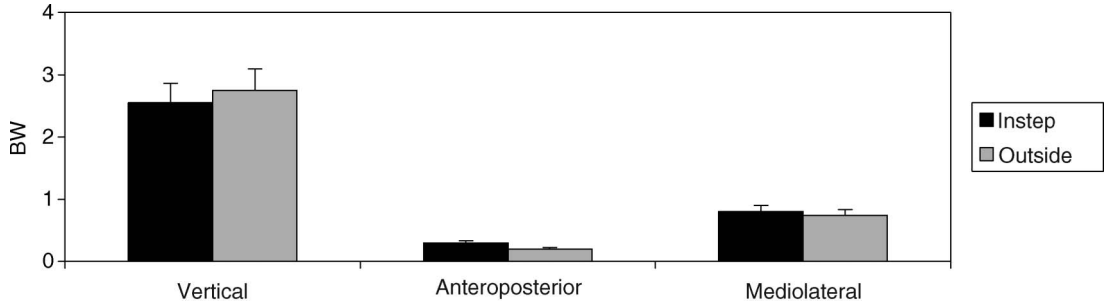

Figure 5. Maximum ground reaction forces (expressed as a percentage of body weight, BW) during each kick in the vertical, anteroposterior, and mediolateral axis.

Ball speed

The speed of the ball is considered a significant biomechanical indicator of kicking success (Kellis & Katis, 2007; Lees & Nolan, 1998). The observed ball speeds for the instep kick are similar to those reported in the literature for instep kicks performed by young soccer players (Luhtanen, 1988). However, to our knowledge, no study has reported ball speeds for the outstep kick, thus making comparisons with other studies difficult.

The ball speed of the outstep kick was significantly lower than that of the instep kick (Table I). Previous studies (Kawamoto, Miyagi, Ohashi, & Fukashiro, 2007; Levanon & Dapena, 1998; Nunome et al., 2002) showed that the instep kick displays higher ball speeds than various other kicking techniques. This was attributed to the development of higher foot speed during the instep kick as opposed to other kicks (Kawamoto et al., 2007; Levanon & Dapena, 1998; Nunome et al., 2002). In the present study, the ball speed/foot speed ratio was significantly higher during the instep kick while non-significant differences in foot speed at contact between the two kicks were observed (Table I). These results suggest that the adjustments made to the foot to contact the ball with the outer portion of the foot reduce foot speed just before contact but not upon contact. Nevertheless, since ball speed differed significantly between the two types of kick, whereas foot speed upon impact did not, it might be suggested that other factors are responsible for the ball speed differences between the two types of kick.

In particular, an important factor in determining the released ball velocity is the quality and the characteristics of foot–ball impact (Asai et al., 2002; Asami & Nolte, 1983). Asami and Nolte (1983) reported that better performance is achieved when the contact point is located closer to the ankle rather than the metatarsals. In this case, the limb becomes more rigid. In contrast, the ''softer'' the collision, the more attenuated the foot–ball impact, which could subsequently reduce the resultant ball speed. Asai et al. (2002) examined instep and inside kicks and reported that when ball impact is achieved with the inside part of the foot and not with the instep portion, the foot slips across the surface of the ball and increases ball spin. A similar type of contact seems to take place when the kick is performed with the outer portion of the foot. Higher ball spin is associated with lower ball speed (Asai et al., 2002). If this is the case, this might explain the lower ball speed observed during the outstep kick (Table I). It could be speculated that during the outstep kicking condition the ''loss'' in ball speed could be compensated by an increase in ball spin, which results in different ball behaviour after release compared with the instep kick.

It has also been suggested that the effective striking mass of the leg affects the velocity of the ball (Lees & Nolan, 1998). Practically this means that a smaller surface of collision would lead to a lower ball speed. This might have also contributed to the lower ball speed observed during the outstep kick, as the orientated foot is likely to be less rigid and thus produce a smaller striking mass. It would appear, therefore, that factors related to foot and ball–foot impact characteristics play a significant role in the released ball velocity between the two kicks.

Joint kinematics

An interesting finding of this study was that no differences were observed in maximum joint and angular segmental velocities in the sagittal plane between the two kicks (Table I). Nunome et al. (2002) also failed to find differences in angular velocity between the instep and the side-foot kick, despite their marked difference in ball speed. The above suggests that both kicks are performed using a similar sagittal movement pattern: the kicking leg rotates backwards and then forwards, following a proximal-to-distal segmental motion.

The present results revealed significant differences in joint displacement values only in the support phase (Figures 1–4). In particular, during the pre-support phase, which coincides with the backswing motion of the kicking leg, both types of kick showed similar sagittal, coronal, and transverse kinematic patterns (Figures 1–4). The swinging limb moved backwards bringing the hip into extension. Simultaneously, the hip was slightly adducted and externally rotated. The knee was flexed and internally rotated, while the ankle was everted and dorsiflexed at the beginning of the motion, but during the latter phases of this backward movement the ankle angle changed orientation.

The support phase was initiated by the forward motion of the hip of the kicking leg (Figure 1). The results showed that, compared with the instep kick, the outstep kick showed a higher hip internal rotation just after the foot landed on the ground followed by a higher hip abduction just before ball impact (Figure 2). This was accompanied by a higher knee internal rotation (Figure 3), although the knee extension–flexion angle was similar for both kicks. Furthermore, the ankle was more inverted for the outstep kick during the entire support phase, as opposed to the eversion position observed for the instep kick upon impact (Figure 4).

The above results show that, compared with the instep kick, the whole lower limb displays higher joint rotations in the transverse and coronal planes, in order to bring the foot to the appropriate position (inverted, dorsiflexed, and supinated) for ball

contact. The absence of any differences in sagittal plane displacements suggests that the outstep kick shows very similar two-dimensional kinematics compared with the instep kick. Nevertheless, the observed intra-kick differences in coronal and transverse joint rotations throughout the forward motion of the kicking leg must be accompanied by exertion of muscle and movement-driven forces that act relative to these joints and produce the observed rotations. One may expect that muscles that control these rotations may also show different activation levels between these two types of kick. For example, Tsaousidis and Zatsiorsky (1996) reported that 50% of the ball speed is imparted to the ball during the collision phase from the energy that is produced from muscle work. The higher inversion of the foot during the outstep kick (Figure 4) may be the result of a different level of muscle activation upon impact during the outstep kick, which requires further examination.

Ground reaction forces and temporal parameters

There were non-significant differences in both temporal values (Table I) and ground reaction forces (Figure 5) between the two types of kick. Although previous studies reported a significant correlation between ground reaction forces and ball speed (Barfield, 1995; Orloff et al., 2008), that was not the case in the present study. Therefore, differences observed in ball speed between the two kicks are likely due to the kinematic differences either during the movement or at impact.

The participants in the present study were pubertal soccer players; they trained for a minimum of 5 years, with a training frequency of twice a week plus one game per week. Thus our results are applicable only to players with the same characteristics. Since the difference between the two types of kicks lies mainly in the way the ball is hit, one would expect similar results in older or more experienced players. However, more research is needed to confirm the above suggestion.

The higher ball speed during the instep kick compared with the outstep kick makes the instep kick a more powerful type of kick. This is in line with previous research comparing the instep kick with other kicking techniques (Asai et al., 2002; Levanon & Dapena, 1998; Nunome et al., 2002). Moreover, it could be suggested that instep kicks are used when high speed and power are needed, whereas outstep kicks should be used when speed needs to be combined with high spin.

A limitation of the present study was the cut-off frequency chosen for filtering the kinematic data based on comparisons between raw and filtered data. A previous study (Nunome, Lake, Georgakis, & Stergioulas, 2006) has shown that this frequency may be rather low given the sampling rate used in the present study. The error due to a lower cut-off is likely to be higher near the foot–ball impact phase and when raw kinematic data are used to calculate variables such as angular acceleration or moments of force or to examine temporal characteristics of calculated kinematic data immediately before or during foot–ball impact, a type of analysis which was not implemented in this study. For these reasons, the selected filtering procedure had a minimal effect on the results of the present study.

Conclusions

Compared with the instep kick, kicking the ball with the outside area of the foot is characterized by higher hip internal rotation, hip abduction, knee internal rotation, and ankle inversion in the phase prior to ball contact. Interestingly, the sagittal plane kinematics, the temporal parameters, and the ground reaction forces were similar between the two kicks. However, the resultant ball speed during the outstep kick was lower compared with that observed during the instep soccer kick. Instep kicking is related to higher ball speed and less knee and hip rotation in the coronal and the transverse plane compared with outstep kicks. It appears that different types of kick require different types of skill training.

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